

HACKATHON PROBLEM STATEMENT

Agentic AI for Logistics Route Optimization

Geospatial Intelligence | Forward & Reverse Logistics | Cost Optimization | Inventory Resilience

Track Agentic AI / Geospatial / Logistics	Level 2nd Year B.E. / B.Tech	Duration 24 Hours	Team Size 3–4 Members
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Problem Overview

Every day, thousands of hardware parts travel across a global network of hubs, satellite locations, and repair centers. When a part fails at a customer site, it needs to travel forward — from a warehouse to the customer. When a part is broken and needs repair, it travels in reverse — from the customer back through the network to a repair center, and once fixed, onward to be redeployed.

This network is expensive and complex. Parts are often routed inefficiently — sent from a distant hub when a closer satellite had stock, or allowed to sit at a congested hub when a faster route was available. SLA commitments are missed, logistics costs balloon, and customers wait longer than necessary.

Your challenge is to build an Agentic AI system, powered by geospatial analysis and historical logistics data, that autonomously analyses routing patterns, identifies inefficiencies, optimizes routing decisions, and produces visual and actionable insights — for both forward and reverse logistics flows.

You are given a dataset of 1,800 real-world-scale logistics transactions across 12 hub locations, 8 repair centers, and 178 unique parts, with a total logistics cost pool of USD 65 million. Your system must make sense of this data and help decision-makers cut costs, reduce delays, and keep repair inventory healthy.

Understanding the Logistics Network

Forward Logistics: Getting Parts to Customers

When a customer's hardware fails and needs a replacement part, the following flow is triggered:

Step	What Happens	Optimization Opportunity
1. Check Destination Nearest Hub	Does the hub closest to the customer have stock?	Use this hub — shortest route, lowest cost

2. Check Satellite Locations	If nearest hub has no stock, check surrounding satellites	Route from satellite if faster than distant primary hub
3. Check Next-Nearest Hub	If satellites are also stocked out, escalate to next hub	Identify which hub to use and optimal routing path
4. Source Internationally	Last resort: source from an international hub	Flag as high-cost; recommend proactive stocking
5. Deliver to Customer	Final leg: part dispatched to customer location	Minimize last-mile cost and transit time

Reverse Logistics: Collecting and Repairing Broken Parts

When a broken part is collected from a customer for repair, it flows in reverse:

Step	What Happens	Optimization Opportunity
1. Collection from Customer	Field engineer collects defective part and packs it	Minimize collection-to-hub transit time
2. Route to Nearest Hub	Part is sent to the nearest regional hub	Consolidate shipments; avoid single-part shipments
3. Hub to Repair Center (TPR)	Hub routes part to the assigned third-party repair center	Choose TPR with capacity; avoid overloaded repair centers
4. Repair & Quality Check	TPR repairs the part; quality verified	Track repair SLA; flag overdue repairs
5. Redeployment	Repaired part returned to inventory for future forward dispatch	Route back to the hub with lowest stock of this part type

The Core Optimization Problem

Right now, routing decisions are often made manually or based on simple rules (e.g., always use the nearest hub).
 Your AI agent must learn from 1,800 historical transactions — analysing cost, SLA performance, hub utilisation, and transit times — to recommend smarter routing decisions.
 The goal: reduce total logistics cost and SLA breach rates by at least 15–20% compared to the current routing patterns visible in the dataset.

Your Dataset

You are provided with a single Excel workbook containing four sheets. All data is synthetic and generated to reflect realistic logistics patterns.

Sheet Name	Contents	Key Fields
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Logistics_Transactions	1,800 rows of historical shipment records covering both forward and reverse flows	Transaction ID, Flow Type, Part No, Origin/Intermediate/Destination Hubs, Quantity, Costs, Dates, SLA Breach, Stock Levels, Tamper Flag, Status
Hub_Location_Master	Details of all 12 hub and satellite locations	Hub ID, City, Country, Latitude, Longitude, Hub Type (Primary/Regional/Satellite/International), Inventory Capacity, Current Stock
TPR_Master	Details of all 8 third-party repair centers	TPR ID, City, Country, Latitude, Longitude, Repair Capacity per Day, Current Workload, SLA Days, Specialisation
Parts_Master	Master catalogue of all 178 parts	Part No, Description, Category, Unit Cost, Weight, Lead Time, Min Stock Level, Fragile, Hazardous flags

Dataset Specs

Total Transactions: 1,800 rows

Total Logistics Cost Pool: USD 65,000,000

Locations: 12 hubs (4 Primary, 4 Regional/Satellite, 4 International) + 8 TPR repair centers across India, Singapore, Malaysia, UAE, Netherlands

Parts: 178 unique SKUs across 8 categories (Compute, Storage, Networking, Power, Display, Cooling, Peripheral, Chassis)

Date Range: Full calendar year 2024

Flow Split: ~60% Forward Logistics, ~40% Reverse Logistics

Coordinates: All hubs and TPRs have real latitude/longitude values for geospatial mapping

Core Challenges to Solve

Challenge 1 — Geospatial Network Visualisation

Before optimising anything, you need to see the network. Build an interactive geospatial map that visualises:

- All hub locations plotted on a map with type indicators (Primary / Regional / Satellite / International)
- All TPR locations plotted with current workload and capacity indicators
- Active shipment routes as lines connecting origin → intermediate hub → destination (colour-coded by flow type and health status)
- A heat-map or bubble chart showing where SLA breaches and tamper alerts are concentrated geographically
- A choropleth or flow diagram showing total logistics cost by origin-destination corridor

Deliverable: An interactive map (use Folium, Plotly, Kepler.gl, Deck.gl, or any mapping library) that a logistics manager can use to understand the network at a glance.

Challenge 2 — Route Efficiency Analysis

Analyse historical routing patterns to identify where inefficiencies exist:

- Identify the top 10 most expensive routing corridors (origin-hub to destination) and explain why they are expensive
- Detect routes where parts were sourced from a distant hub even though a closer hub or satellite had sufficient stock
- Calculate the average cost per unit per km for each corridor and identify outliers
- Identify which hubs are consistently over-utilised (high-cost routing bottlenecks) vs under-utilised
- Compare actual transit days vs expected for each corridor and flag chronic underperformers

Deliverable: A visual dashboard showing corridor efficiency rankings, cost-per-km analysis, and hub utilisation heatmap.

Challenge 3 — Agentic Routing Recommender

Build an AI agent that, given a new part request, recommends the optimal routing path. The agent must:

- Accept inputs: Part No, Quantity, Priority, Destination City
- Query the Hub Location Master and current stock levels to determine which hubs have available stock
- Apply a routing decision tree: Nearest Hub with Stock → Nearest Satellite → Next Hub → International Source
- Calculate the estimated cost and transit time for each candidate route
- Recommend the best route with a justification (cost, distance, availability, SLA confidence)
- Generate an alternative route as a backup if the primary route has risk (low stock, overloaded hub)
- Learn from historical SLA breach patterns: if a route has a 40%+ breach history, warn the user and downrank it

Deliverable: A working agent (demo via CLI, Streamlit, or Gradio) that accepts a routing request and outputs a ranked recommendation with cost estimates and risk flags.

Challenge 4 — Cost Optimisation Model

Your agent should be able to answer: "How much could we save if we routed smarter?"

- Identify all transactions in the dataset where a suboptimal route was used (i.e., a cheaper alternative was available)

- Calculate the excess cost of the suboptimal routing for each such transaction
- Aggregate potential savings by hub, by part category, by logistics partner, and by flow type
- Model the impact of holding more inventory at satellite locations (reducing the need for expensive inter-hub transfers)
- Produce a "What-If" analysis: if the top-3 most expensive corridors were re-routed optimally, what is the total saving?

Deliverable: A cost optimisation report (visual + tabular) showing current vs optimised cost per corridor, total potential savings, and inventory investment needed to unlock those savings.

Challenge 5 — Reverse Logistics Optimisation

Optimising the reverse flow (broken part collection → repair → redeployment) is equally important:

- Analyse which repair centers (TPRs) are over-capacity and which are underutilised
- Identify broken parts that were routed to a distant TPR when a closer one with capacity and specialisation match was available
- Calculate the cost of redeployment: once a part is repaired, is it being sent back to the hub with the highest demand for that part?
- Detect repair consolidation opportunities: multiple small single-part shipments from the same hub to the same TPR that could be batched
- Flag TPRs where current stock of repair inputs (spare components) is below minimum threshold — generate a restock alert

Deliverable: A reverse logistics optimisation dashboard with TPR utilisation view, consolidation opportunities, redeployment efficiency analysis, and restock alert list.

Challenge 6 — SLA Breach Predictor

Train a lightweight predictive model to predict whether a shipment will breach its SLA before it is dispatched:

- Features to consider: origin hub, destination, part category, priority, quantity, logistics partner, departure day of week, current hub utilisation, stock level at origin
- Target variable: SLA_Breach (YES / NO) from the Logistics_Transactions sheet
- Build a simple classifier (Decision Tree, Random Forest, or Logistic Regression — accessible to 2nd year students)
- Evaluate performance: accuracy, precision, recall, confusion matrix
- Integrate the predictor into the routing recommender: if a recommended route has a predicted SLA breach probability > 50%, flag it

Deliverable: A trained model with evaluation metrics and integration into the routing agent as a risk signal.

Reference Architecture

Your solution should be structured as a set of cooperating agents, each handling a distinct domain. Any cloud platform or framework is permitted.

Agent / Component	Responsibility
Data Ingestion Agent	Loads the Excel dataset; validates schema; normalises dates and numeric fields; flags data quality issues
Geospatial Mapper	Renders the network map; plots hubs, TPRs, and active routes; generates heatmaps and corridor flow diagrams
Route Analyser Agent	Analyses historical routes; calculates cost per km; identifies suboptimal corridors; benchmarks hub utilisation
Routing Recommender Agent	Accepts new shipment requests; applies decision tree logic; scores candidate routes; outputs ranked recommendations with costs
Cost Optimisation Agent	Quantifies excess cost from suboptimal routing; models inventory investment trade-offs; generates What-If savings analysis
Reverse Logistics Agent	Analyses reverse flow; TPR capacity and utilisation; consolidation opportunities; redeployment routing; restock alerts
SLA Predictor	Predicts SLA breach probability for a proposed route; integrates as a risk flag into the routing recommender
Insight & Report Generator	Produces all visualisations, dashboards, and narrative summaries; formats outputs for logistics managers
Orchestrator / Planner	Coordinates agent handoffs; manages context flow; ensures agents share common data state

Mandatory Outputs

Your solution must produce the following outputs for evaluation:

Output	Description
Network Map	Interactive geospatial map of all hubs, TPRs, and active routes with cost and status overlays
Corridor Efficiency Report	Ranked list of top inefficient corridors with cost-per-km, breach rate, and improvement potential
Routing Recommendation	Live agent demo: input a part request, output an optimised route with cost, ETA, and risk score
Cost Optimisation Report	Current vs optimised cost comparison; total potential savings; What-If analysis for top corridors

Reverse Logistics Dashboard	TPR utilisation view; consolidation opportunities; redeployment efficiency; restock alerts
SLA Breach Predictor	Trained classifier with evaluation metrics; integrated risk flag in routing recommendations
Executive Summary Slide	One-slide portfolio view: network health, top cost insights, potential savings, and top recommendations

Sample Insights to Discover

Here are the types of findings your agent should be able to surface from the dataset. These are examples — your analysis may find others:

Insight Type	Example Finding
Routing Inefficiency	23% of forward logistics shipments were sourced from an international hub even though a domestic satellite had stock — average excess cost: USD 4,200 per shipment
Hub Bottleneck	Mumbai Hub processed 34% of all reverse logistics transactions but only has 18% of total network capacity — contributing to a 41% SLA breach rate on that corridor
Consolidation Opportunity	84 separate single-unit shipments from Bangalore Hub to TPR-BLR-01 over Q2 could have been batched into 12 consolidated shipments — estimated saving: USD 180,000
SLA Pattern	Shipments dispatched on Fridays have a 2.3x higher breach rate than midweek dispatches — a logistics partner scheduling pattern
Redeployment Gap	Repaired Compute parts from TPR-CHE-01 are being sent back to Chennai Hub, which already has excess stock — while Kolkata Satellite is critically understocked for the same part type
Cost Corridor	The Delhi → Singapore corridor has the highest per-unit logistics cost in the dataset (avg USD 340/unit) but accounts for only P3/P4 priority shipments — a clear candidate for sea freight re-routing

Suggested Participant Workflow

1. Load the Excel dataset and explore all four sheets; validate data types, check for nulls, and compute basic summary statistics.
2. Build the geospatial map with all hub and TPR locations plotted; add at least 20 sample shipment routes as lines.
3. Run corridor efficiency analysis: compute cost per km, average transit days, and SLA breach rate for each origin-destination pair.

4. Build the routing recommender agent: implement the 4-step decision tree and demonstrate it with 3 live routing queries.
5. Run cost optimisation analysis: identify the top 5 suboptimal corridors and quantify total potential savings.
6. Analyse the reverse logistics flow: identify consolidation opportunities and TPR utilisation imbalances.
7. Train the SLA breach predictor and integrate it as a risk flag into the routing agent.
8. Produce the executive summary slide and all mandatory outputs; prepare a 5-minute live demo.

Suggested Technology Stack

Teams are free to use any tools, frameworks, or cloud platforms. Here are suggestions:

Domain	Suggested Options
Data Processing	Python with pandas, numpy; Jupyter notebooks; Google Colab for free GPU/compute
Geospatial Mapping	Folium (easiest for beginners), Plotly Express, Kepler.gl, Deck.gl, or Leaflet.js
Geospatial Analysis	GeoPandas, Shapely, OSMnx for distance calculations; Haversine formula for simple lat/lon distance
Agent Framework	LangChain, LangGraph, CrewAI, or custom Python orchestration with LLM APIs
LLM for Reasoning	OpenAI GPT-4o, Anthropic Claude, or Google Gemini to generate natural-language routing rationales
ML / Prediction	scikit-learn for SLA predictor (RandomForest, LogisticRegression); XGBoost for stronger performance
Visualisation	Matplotlib, Seaborn, Plotly for charts; Streamlit or Gradio for interactive demo UI
Cloud Platform	Any cloud — AWS, Azure, GCP; or fully local; Streamlit Cloud for free hosting of demos
Routing/Distance	Google Maps Distance Matrix API, OpenRouteService (free), or simple Haversine-based distance

Bonus Challenges (Optional)

- Dynamic Re-routing: If a hub goes offline mid-transit (simulate by removing a hub from the active list), automatically re-route all affected in-flight shipments

- Carbon Footprint Calculator: Estimate CO2 emissions per shipment based on distance and transport mode; include a carbon-optimised routing option
- Inventory Balancing Agent: Continuously rebalance stock across hubs using historical demand forecasts to pre-position parts before requests arrive
- Natural Language Query Interface: Let users ask questions in plain English — e.g., "Which hub has the most SLA breaches for Storage parts?" — and get a visual answer
- Live Route Tracker Simulation: Simulate a shipment moving in real time across the map, with status updates at each hub scan point
- Multi-objective Optimisation: Allow the user to trade off cost vs. speed vs. carbon — and show the Pareto frontier of optimal solutions

Judging Rubric (Mentor-Scored)

Mentors will assess submissions based on the following 13 dimensions. Winners will be selected solely based on mentor rankings.

#	Criterion	What Mentors Look For
1	Solution Quality	Effectiveness and completeness of the proposed solution
2	Tool Stack Used	Appropriateness and sophistication of technologies applied
3	Presentation & Pitch	Clarity, engagement, and delivery of the final presentation
4	Feasibility & Integration	Practicality and ease of integration into real-world scenarios
5	Innovation & Originality	Novelty of the idea and creative problem-solving
6	Modularity & Reusability	Scalability and adaptability of the solution components
7	Impact Potential	Anticipated value and influence of the solution
8	Testing & Validation	Evidence of robustness and reliability through testing
9	User Experience & Design	Intuitiveness, accessibility, and visual appeal
10	Documentation Clarity	Quality and completeness of supporting documentation
11	Security & Privacy	Consideration of data protection and secure practices
12	Explainability & Transparency	Clarity in how the solution works and makes decisions
13	Feedback Loop / Learning Capability	Ability to adapt, learn, or improve over time

Note on Winner Selection

Winners will be selected solely based on mentor rankings across all 13 dimensions above. There is no audience vote or self-scoring. Mentor decisions are final.

Security & Privacy Considerations

Even at prototype level, keep these principles in mind:

- Treat location coordinates and shipment volumes as sensitive operational data — do not expose them in public URLs or logs
- Parameterise all thresholds (SLA days, cost limits, stock minimums) — never hard-code business rules
- Maintain an audit trail of every recommendation your agent makes, with the data and rules used as evidence
- If integrating with external APIs (mapping, LLMs), ensure API keys are stored in environment variables, never in source code

USD 65 million in logistics cost. 1,800 shipments. 12 hubs. Your agent's job: make every route count.

Questions? Reach out to your hackathon coordinators.